



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

**Salem Terminal TO J.P.RAMESH AGENCIES IOC
DEALER**

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

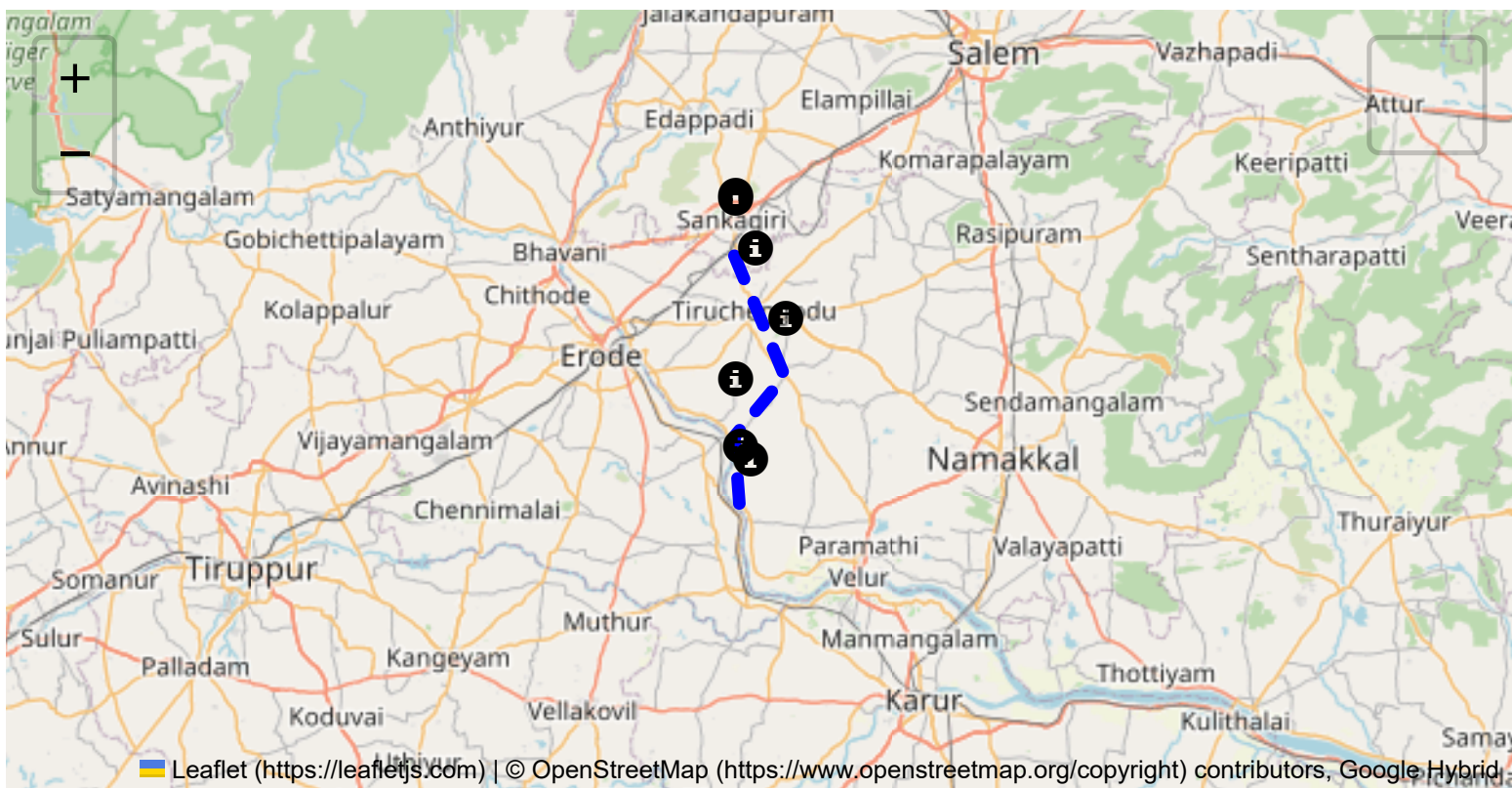
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

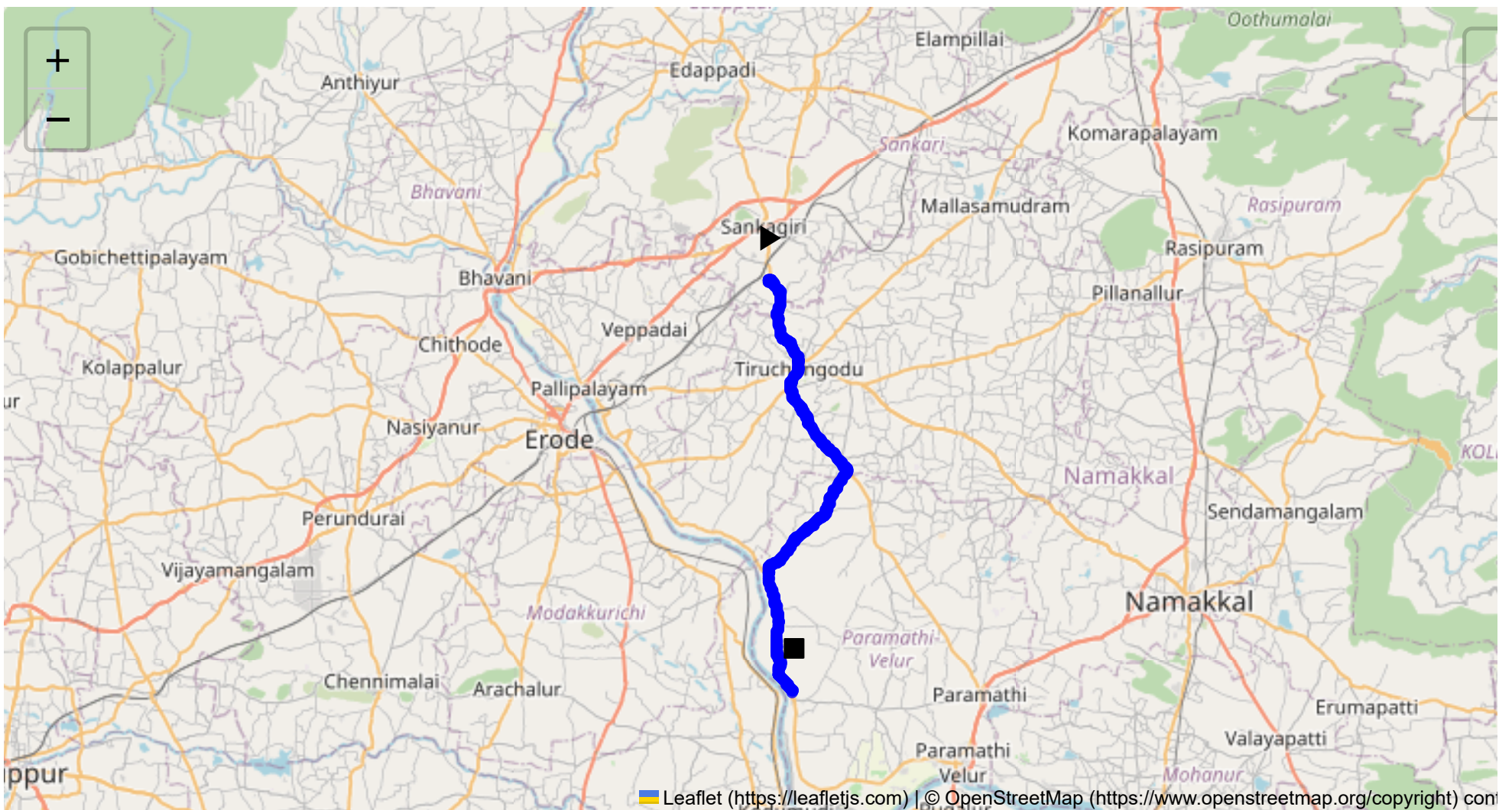
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 37.44 km
Estimated Duration: 0.9 hours
Adjusted Duration (Heavy Vehicle): 1.1 hours
Start: (11.4381, 77.8734)
End: (11.157407, 77.89016)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route begins at CVQF+23W in Sangagiri, Tamil Nadu, and ends at 5V4R+W3J in Jedarpalayam, Tamil Nadu. It passes through several notable points, including Puthur, Tiruchengode, Sullipalayam, Solasiramani, and Kothamangalam. This route spans approximately 37.44 kilometers and winds through both rural and semi-urban landscapes.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The region typically experiences a tropical climate. The summer months (March to June) can be very hot, leading to possible vehicle overheating. The monsoon season occurs from June to September, causing potential road flooding and reduced visibility due to heavy rains. During the northeast monsoon (October to December), torrential rains can create hazardous driving conditions and may lead to landslides or flooding in low-lying areas.

3. Analysis of Traffic Patterns

The traffic in this area peaks during morning (7:00 AM - 9:00 AM) and evening (5:00 PM - 8:00 PM) due to local commuting patterns. Congestion is particularly notable near schools, such as near Girls High School and marketplaces in Tiruchengode. Traffic is generally lighter in rural parts of the route but can be unpredictable near villages due to agricultural vehicles and pedestrians.

4. Assessment of Road Quality and Infrastructure

The road quality varies along this route. While the main roads are generally well-maintained, some rural stretches may feature narrower lanes, potholes, and uneven surfaces. The infrastructure near Tiruchengode is better developed, but attention should be paid to road conditions in more rural segments, especially after heavy rains.

5. Suggestions for Alternative Routes for Emergencies

In case of emergencies, one could consider taking the NH79 as an alternative to reach closer major towns for assistance. However, specific alternatives should be planned depending on the exact location and nature of the emergency, consulting local traffic advisories.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu has strict regulations regarding the transport of hazardous materials. This includes specific permissible routes, timing restrictions (primarily to avoid school and market hours), and requirements for vehicle safety and driver certification. Transporters must have the necessary permits and adhere to these regulations to avoid penalties.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

While specific historical data may not be published readily, incidents involving heavy vehicles are not uncommon in heavily trafficked areas like Tiruchengode. These generally involve rollovers or collisions due to road conditions and sharp turns. Increased caution is advised after heavy rainfall, which can exacerbate these risks.

8. Environmental Considerations and Sensitive Areas

The route passes near agricultural lands and small water bodies which are sensitive to contamination. Care must be taken to prevent any spills, especially when transporting hazardous materials. There are also school zones which require careful navigation and reduced speed.

9. Analysis of Communication Coverage

Communication coverage is generally reliable along the main roads, but there may be occasional dead zones in rural segments, especially between Puthur and Solasiramani. It is advisable to check mobile network strength in advance and prepare for potential communication interruptions.

10. Estimated Emergency Response Times

Emergency response times can vary considerably. In urban areas such as Tiruchengode, response times may be around 15-30 minutes due to nearby services. In more rural areas, this can extend to an hour or more, depending on accessibility and road conditions.

11. Overall Summary of Risk Assessment

This route presents several potential hazards, especially related to weather conditions, road quality, and traffic patterns. While parts of the road pass through well-maintained areas, rural segments may pose challenges, particularly for heavy and hazardous material transport. Careful planning and adherence to safety regulations are essential to mitigate risk. Emergency plans and communication strategies should be in place due to varying response times and coverage areas. Transporters must be vigilant, especially during adverse weather conditions, and comply strictly with local hazardous material transport regulations.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43961, 77.87341	30 KM/Hr
3	Turn	Medium	11.43968, 77.87345	30 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	High	11.37868, 77.89498	15 KM/Hr
6	Turn	High	11.37850, 77.89453	15 KM/Hr
7	Turn	High	11.30859, 77.92918	15 KM/Hr
8	Turn	Medium	11.23924, 77.87368	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
7	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37868, 77.89498



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.30859, 77.92918



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.23924, 77.87368
